

Advanced Time Series Analysis and Forecasting

Seminar 4: Cointegration revisited: VECM, ARDL and panel data



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Acronyms used in this material (1/2)

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARDL**: AutoRegressive Distributed Lag (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BUBOR**: Budapest Interbank Offered Rate
- **CADF**: Cross-sectionally augmented Dickey–Fuller (regression)
- **CD**: Cross-section Dependence (test of Pesaran)
- **CIPS**: Cross-sectionally augmented IPS (test)
- **ECM**: Error Correction Model
- **EU**: European Union
- **EU-27**: the 27 member states of the European Union
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices

Acronyms used in this material (2/2)

- **HQ**: Hannan–Quinn (information criterion)
- **IFS**: International Financial Statistics (IMF)
- **IMF**: International Monetary Fund
- **IPS**: Im–Pesaran–Shin (panel unit-root test)
- **KPSW**: King–Plosser–Stock–Watson (common-trends model, 1991)
- **LLC**: Levin–Lin–Chu (panel unit-root test)
- **LR**: Likelihood Ratio (test)
- **MG**: Mean Group (estimator)
- **NARDL**: Nonlinear AutoRegressive Distributed Lag (model)
- **PMG**: Pooled Mean Group (estimator)
- **PSS**: Pesaran–Shin–Smith
- **RA**: Reinsel–Ahn (small-sample correction)
- **SE**: Standard Error
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax
- **VECM**: Vector Error Correction Model

Today's question and seminar structure

- The question of the seminar
  - **how do we decide whether non-stationary series share a long-run equilibrium, test what theory says about it, and keep the answer honest in small samples and across countries?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the trace test from eigenvalues; degrees of freedom of LR tests	Solved, Proposed	A1
A3, A4	the Granger representation; the permanent shock of a common-trends model	Solved, Proposed	A3
A5, A6	an ARDL in error-correction form; a bounds-test decision	Solved, Proposed	A5
A7, A8	the CD statistic; the Hausman test of PMG against MG	Solved, Proposed	A7
B1, B2	Hungarian pass-through: rank tests; tests on β and α	Solved, Proposed	B1
B3, B4	bounds test for fuel prices; the KPSW common trend on longer samples	Solved, Proposed	B3
B5	EU-27 inflation: IPS against CIPS	Proposed	A7, B1
C1, C2	asymmetric fuel-price adjustment; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; its solution is discussed in the seminar
- **A**: calculations and derivations on paper; **B**: real data, solved in the notebook, each closing with an interpretation question; **C**: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
Hungary: lending and deposit rates; BUBOR 3M	IMF IFS; Eurostat	monthly	2005–2026
Romania: HICP fuels; Brent; USD/RON	Eurostat; FRED; BNR	monthly	2007–2026
US consumption, investment, GDP	FRED	quarterly	1949–2025
EU-27 HICP inflation	Eurostat	annual	1999–2025

- In the notebook: `johansen`, `restricted_ml`, `weak_exog_lr`, `boot_trace`, `ardl_ecm`, `bounds_quantiles`, `cd_test`, `ips_tbar`, `cips`; no account or key is needed

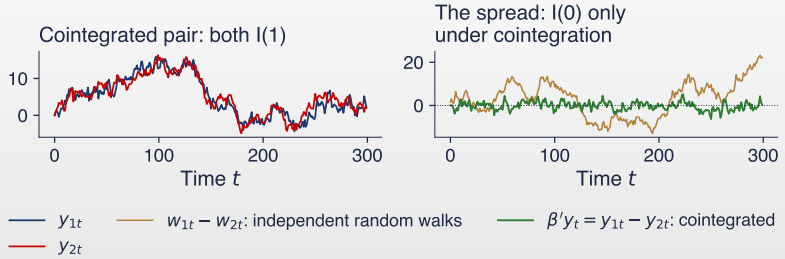
Prerequisites for Today (1/26): integration and cointegration

- **Order of integration**: y_t is I(d) if $\Delta^d y_t$ is stationary and $\Delta^{d-1} y_t$ is not
  - $\Delta y_t = y_t - y_{t-1}$: the first difference; Δ^d: d successive differences
  - I(1) example: the random walk $y_t = y_{t-1} + \varepsilon_t$, with $\text{Var}(y_t) = t\sigma^2$ growing without bound
- **Cointegration** Engle and Granger (1987): all components of $y_t \in \mathbb{R}^n$ are I(1), but r linear combinations are I(0)

$$y_t \sim \text{I}(1), \qquad \beta' y_t \sim \text{I}(0), \qquad \beta \in \mathbb{R}^{n \times r}$$

  - β: the cointegrating vectors; $\beta' y_t$: the deviations from the long-run equilibrium, which return to their mean
  - r: the cointegration rank, $0 \leq r < n$; $n - r$: the number of common stochastic trends
- A regression between independent I(1) series is spurious
  - its R^2 and t-ratios do not have their usual distributions: cointegration must be tested

Prerequisites for Today (2/26): a simulated cointegrated pair



- Left: two simulated I(1) series that wander together; VECM with $\alpha = (-0.25, 0.10)'$, $\beta = (1, -1)'$ (next frames)
- Right: the spread $\beta' y_t = y_{1t} - y_{2t}$ keeps returning to zero (standard deviation 1.9)
  - the difference of two independent random walks wanders away (standard deviation 8.1): no equilibrium

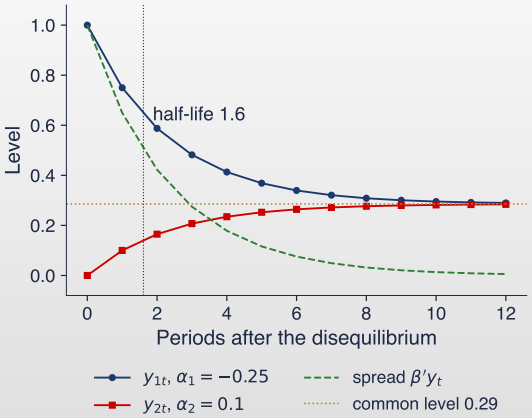
Prerequisites for Today (3/26): the VECM

- VECM (TSA, Chapter 7): the change of each variable reacts to last period's deviations from equilibrium and to past changes

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t, \qquad \Pi = \alpha\beta'$$

- Notation
  - $y_t \in \mathbb{R}^n$: the n I(1) variables; p: the lag order of the VAR in levels; T: the sample size
  - β $(n \times r)$: the cointegrating vectors; α $(n \times r)$: the adjustment coefficients (loadings)
  - Γ_i $(n \times n)$: short-run matrices; D_t: deterministic terms (constant, trend) with coefficients Φ
  - $\varepsilon_t \sim N(0, \Omega)$: the innovations, i.i.d., with covariance matrix Ω
- The rank of Π is the cointegration rank r
  - $r = 0$: $\Pi = 0$, a VAR in differences; $r = n$: Π of full rank, a stationary VAR in levels
  - $H(r)$: the hypothesis that the rank is at most r, i.e. $\Pi = \alpha\beta'$ with α, β of r columns

Prerequisites for Today (4/26): error correction



- Bivariate VECM without lags, $\beta = (1, -1)'$

$$\Delta y_{it} = \alpha_i (y_{\mathbf{1},t-\mathbf{1}} - y_{\mathbf{2},t-\mathbf{1}}) + \varepsilon_{it}$$

  - $\alpha_{\mathbf{1}} < 0$: $y_{\mathbf{1}}$ falls when it is above the equilibrium
  - $\alpha_{\mathbf{2}} > 0$: $y_{\mathbf{2}}$ rises towards $y_{\mathbf{1}}$
  - $\alpha_i = 0$: variable i does not adjust

- The spread is an AR(1)

$$\beta' y_t = (1 + \beta'\alpha)\, \beta' y_{t-\mathbf{1}} + \beta' \varepsilon_t$$

  - chart: $1 + \beta'\alpha = 1 - 0.25 - 0.10 = 0.65$
  - half-life $\ln 0.5 / \ln 0.65 = 1.6$ periods
  - both variables settle at a common level (0.29)

Prerequisites for Today (5/26): Johansen's reduced-rank regression

- Step 1 Johansen (1991): regress Δy_t and y_{t-1} on $\Delta y_{t-1}, \ldots, \Delta y_{t-p+1}$ and D_t
  - R_{0t}, R_{1t}: the two residual vectors, i.e. changes and levels purged of the short run
- Step 2: the moment matrices

$$S_{ij} = \frac{1}{T} \sum_{t=1}^{T} R_{it} R_{jt}', \qquad i, j \in \{0, 1\}$$

- Step 3: the eigenvalues $1 > \hat{\lambda}_1 > \cdots > \hat{\lambda}_n > 0$ solve

$$\left| \lambda S_{11} - S_{10} S_{00}^{-1} S_{01} \right| = 0$$

  - $|\cdot|$: the determinant; $\hat{\lambda}_i$: the squared canonical correlations between R_{0t} and R_{1t}
  - a large $\hat{\lambda}_i$: a combination of levels that predicts the changes, i.e. a cointegrating relation; its eigenvector is a column of $\hat{\beta}$
  - maximised log-likelihood under $H(r)$: $\ln L_{\max}(r) = c - \frac{T}{2} \sum_{i=1}^{r} \ln(1 - \hat{\lambda}_i)$, c a constant

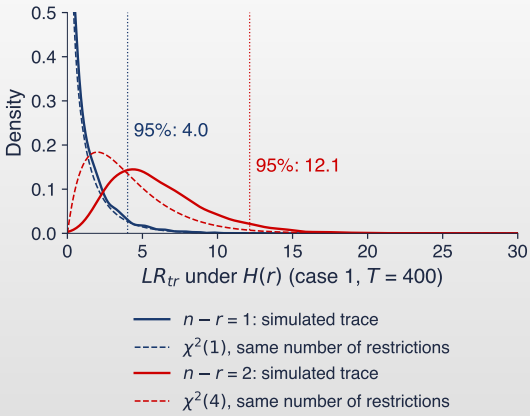
Prerequisites for Today (6/26): the trace and max-eigenvalue tests

- Two likelihood-ratio statistics, from twice the gain in $\ln L_{\max}$

$$LR_{tr}(r) = -T \sum_{i=r+1}^{n} \ln(1-\hat{\lambda}_i), \qquad LR_{\max}(r) = -T \ln(1-\hat{\lambda}_{r+1})$$

- Hypotheses
  - trace: $H(r)$ against $H(n)$, the stationary VAR; max-eigenvalue: $H(r)$ against $H(r+1)$
  - both are large when the eigenvalues that $H(r)$ sets to zero are not small
- Sequential procedure
  - test $r = 0, 1, \ldots, n-1$ in turn; $\hat{r}$ = the first r that is not rejected
- Reference distribution: non-standard (functionals of Brownian motion)
  - simulated critical values MacKinnon, Haug and Michelis (1999), which depend on $n-r$ and on the deterministic case
  - reject $H(r)$ when the statistic exceeds the simulated 95% quantile

Prerequisites for Today (7/26): the null distribution of the trace statistic



- Simulated $LR_{tr}(r)$ under $H(r)$
  - $n - r$ independent random walks, case 1 (no deterministic terms), $T = 400$
- 95% quantiles: 4.0 ($n - r = 1$) and 12.1 ($n - r = 2$)
  - $H(r)$ imposes $(n - r)^2$ restrictions on Π
  - the χ^2 quantiles with these degrees of freedom are 3.84 and 9.49: too small
- The gap grows with $n - r$: always use the simulated tables of the right case

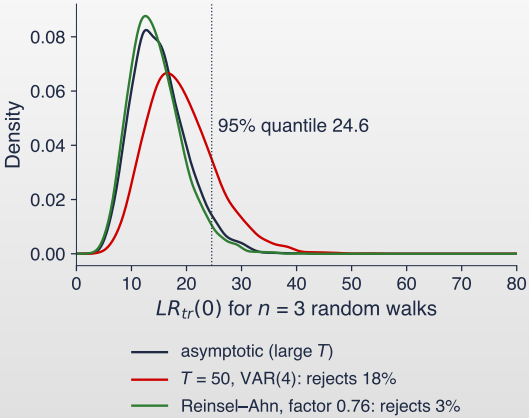
Prerequisites for Today (8/26): the five deterministic cases

- General form, constant and trend split between the relation and the changes

$$\Delta y_t = \alpha(\beta' y_{t-1} + \rho_0 + \rho_1 t) + \gamma_0 + \gamma_1 t + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \varepsilon_t$$

- ρ_0, ρ_1 ($r \times 1$): constant and trend inside $\beta' y$; γ_0, γ_1 ($n \times 1$): outside it
  - case 1: no deterministic terms; case 2: only ρ_0, so the levels have no drift
  - case 3: unrestricted constant γ_0: linear trends in the levels, none in $\beta' y$
  - case 4: γ_0 and ρ_1: a trend restricted to $\beta' y$ (trend-stationary equilibrium)
  - case 5: unrestricted trend γ_1: quadratic trends in the levels
- Choose the case from the shape of the data and from economics, before testing
  - each case has its own critical values for the rank tests

Prerequisites for Today (9/26): small samples: the Reinsel–Ahn factor



- Reinsel–Ahn correction Reinsel and Ahn (1992)

$$LR_{tr}^{RA}(r) = \frac{T - np}{T} LR_{tr}(r)$$

  - np: lag coefficients per equation of the VAR(p) in levels
  - a degrees-of-freedom factor below 1: it shrinks the statistic
- Chart: $n = 3$ independent random walks, VAR(4), $T = 50$
  - the asymptotic test rejects the true $H(0)$ in 18% of samples, not 5%
  - with the factor 0.76 it rejects in 3%
- Too many lags for the sample size make the asymptotic rank too large

Prerequisites for Today (10/26): the wild bootstrap and the lag order

- Wild bootstrap of the rank test Cavaliere, Rahbek and Taylor (2012)
  - estimate the VECM under $H(r)$ and keep its residuals $\hat{\varepsilon}_t$
  - draw $\varepsilon_t^* = \hat{\varepsilon}_t w_t$, $w_t \sim N(0,1)$ i.i.d.; rebuild y_t^* recursively from the restricted model
  - the bootstrap data have $n - r$ unit roots by construction; recompute LR_b^*, $b = 1, \ldots, B$
- Bootstrap p-value

$$p^* = \frac{1}{B} \sum_{b=1}^{B} 1\{LR_b^* \geq LR_{tr}(r)\}$$

  - $1\{\cdot\}$: 1 if the condition holds, 0 otherwise; the weights w_t keep any heteroskedasticity of $\hat{\varepsilon}_t$
- Lag order of the VAR in levels by an information criterion

$$IC(p) = \ln\left|\hat{\Sigma}_p\right| + c_T \frac{pn^2}{T}$$

  - $\hat{\Sigma}_p$: residual covariance matrix of the VAR(p); pn^2: the number of lag coefficients
  - AIC: $c_T = 2$; BIC: $c_T = \ln T$; HQ: $c_T = 2 \ln \ln T$; choose the p with the smallest $IC(p)$

Prerequisites for Today (11/26): identification and restrictions on β

- Only the space spanned by β is identified: $\alpha\beta' = (\alpha Q^{-1\prime})(\beta Q)'$ for any invertible Q $(r \times r)$
  - just-identification: $r-1$ restrictions per vector besides the normalisation, $r(r-1)$ in total
  - they only choose a basis of the same space: the likelihood does not change
- The same restriction on all vectors

$$\beta = H\varphi, \qquad \text{degrees of freedom } r(n_1 - s)$$

  - n_1: rows of β, i.e. the variables plus the restricted deterministic terms
  - H $(n_1 \times s)$: known; φ $(s \times r)$: free; example: excluding one variable gives $s = n_1 - 1$
- All r vectors fully known from theory
  - degrees of freedom $r(n_1 - r)$: the r normalisations and the $r(r-1)$ identifying restrictions are free

Prerequisites for Today (12/26): restrictions on α and weak exogeneity

- Restrictions on the adjustment coefficients

$$\alpha = A\psi, \qquad \text{degrees of freedom } r(n-m)$$

  - A $(n \times m)$: known; ψ $(m \times r)$: free
- **Weak exogeneity** of x_t: its row of α is zero ($m = n - 1$, so r restrictions)
  - x_t does not adjust to disequilibria; the other equations, conditional on Δx_t, carry all the information on β
  - so a single-equation (conditional) model of the other variables is valid
- LR test of restrictions on α or β

$$LR = T \sum_{i=1}^{r} \ln \frac{1-\tilde{\lambda}_i}{1-\hat{\lambda}_i} \xrightarrow{d} \chi^2(\text{df})$$

  - $\tilde{\lambda}_i$: the eigenvalues of the restricted problem; df: the degrees of freedom above
  - unlike the rank tests, a standard χ^2 reference once r is fixed

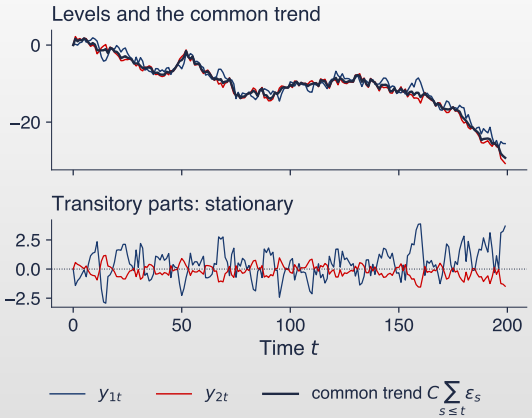
Prerequisites for Today (13/26): the Granger representation

- The levels are random-walk trends plus stationary terms

$$y_t = C\sum_{s=1}^{t}\varepsilon_s + C^*(L)\varepsilon_t + y_0^*, \qquad C = \beta_\perp(\alpha_\perp'\Gamma\beta_\perp)^{-1}\alpha_\perp'$$

- Notation
  - $\Gamma = I_n - \sum_{i=1}^{p-1}\Gamma_i$; $\alpha_\perp$, $\beta_\perp$ $(n \times (n-r))$: orthogonal complements, $\alpha'\alpha_\perp = 0$, $\beta'\beta_\perp = 0$
  - $C^*(L)\varepsilon_t$: a stationary moving average; y_0^*: depends on the initial values
- Reading C
  - C: the long-run effect of a shock ε_t on the levels; $\beta' C = 0$, so $\beta' y_t$ has no stochastic trend
  - the $n-r$ common trends are $\alpha_\perp'\sum_s \varepsilon_s$: shocks to variables that adjust little weigh most
  - $n = 2$, $\beta = (1, -1)'$: $\beta_\perp = (1, 1)'$ and $\alpha_\perp = (\alpha_2, -\alpha_1)'$

Prerequisites for Today (14/26): the common trend of a simulated VECM



- VECM of the error-correction chart: $\alpha = (-0.25, 0.10)'$, $\Gamma = I$
  - $\alpha_\perp = (0.10, 0.25)'$, $\alpha_\perp'\beta_\perp = 0.35$
  - each row of C is $(0.29, 0.71)$
- Top: both levels follow the same trend $C\sum_s \varepsilon_s$
  - y_2 adjusts less ($|\alpha_2| < |\alpha_1|$): its shocks weigh 0.71 in the trend
- Bottom: $y_t - C\sum_s \varepsilon_s$ is stationary

Prerequisites for Today (15/26): the permanent shock and its variance share

- One common trend ($n - r = 1$): the KPSW permanent shock King, Plosser, Stock and Watson (1991)

$$u_t^P = \frac{\alpha_\perp' \varepsilon_t}{(\alpha_\perp' \Omega \alpha_\perp)^{1/2}}, \qquad b^P = \frac{\Omega \alpha_\perp}{(\alpha_\perp' \Omega \alpha_\perp)^{1/2}}$$

  - u_t^P: the innovation of the common trend, scaled to unit variance
  - $b^P = \mathrm{Cov}(\varepsilon_t, u_t^P)$: the effect of u_t^P on the n variables on impact; Cb^P: its long-run effect

- Share of u^P in the h-step forecast error variance of variable i

$$s_i(h) = \frac{\sum_{j=0}^{h-1} (\Theta_j b^P)_i^2}{\sum_{j=0}^{h-1} (\Theta_j \Omega \Theta_j')_{ii}}$$

  - Θ_j: the moving-average (impulse-response) matrices of the levels, $\Theta_0 = I_n$, $\Theta_j \to C$
  - $h = 1$: $s_i(1) = (b_i^P)^2 / \Omega_{ii}$; a share between 0 and 1

Prerequisites for Today (16/26): ARDL and its error-correction form

- ARDL(p, q): y_t on its own p lags and on x_t with q lags

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=0}^{q} \beta_j' x_{t-j} + u_t$$

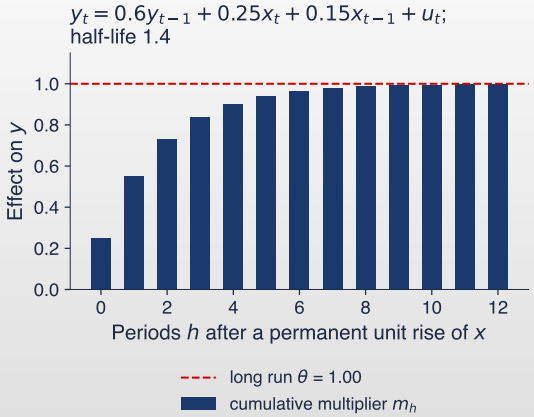
  - x_t: k regressors; c: intercept; ϕ_i, β_j: coefficients; u_t: error

- The same model in conditional ECM form

$$\Delta y_t = c + \pi_{yy} y_{t-1} + \pi_{yx}' x_{t-1} + \sum_{i=1}^{p-1} \psi_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \omega_j' \Delta x_{t-j} + u_t$$

  - $\pi_{yy} = \sum_i \phi_i - 1$: the speed of adjustment, negative if y returns to equilibrium; $\pi_{yx} = \sum_j \beta_j$
  - ψ_i, ω_j: the short-run coefficients
  - long run $\theta = -\pi_{yx}/\pi_{yy}$: the equilibrium is $y = \theta' x$ plus a constant
  - half-life of a disequilibrium in an ARDL(1, q): $\ln 0.5 / \ln(1 + \pi_{yy})$ periods

Prerequisites for Today (17/26): dynamic multipliers

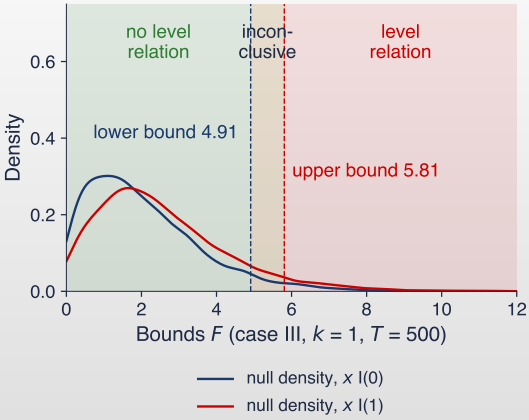


- Cumulative multiplier: the effect on y_{t+h} of a permanent unit rise of x from t on
  - ARDL(1, 1): $m_0 = \beta_0$, $m_h = \phi_1 m_{h-1} + \beta_0 + \beta_1$
  - $m_h \to \theta$ geometrically, at the rate ϕ_1
- Chart: $\phi_1 = 0.6$, $\beta_0 = 0.25$, $\beta_1 = 0.15$
  - $\pi_{yy} = -0.4$, $\pi_{yx} = 0.4$, so $\theta = 1$
  - impact 0.25; 73% of the long run after 2 periods
  - half-life $\ln 0.5 / \ln 0.6 = 1.4$ periods

Prerequisites for Today (18/26): the bounds test

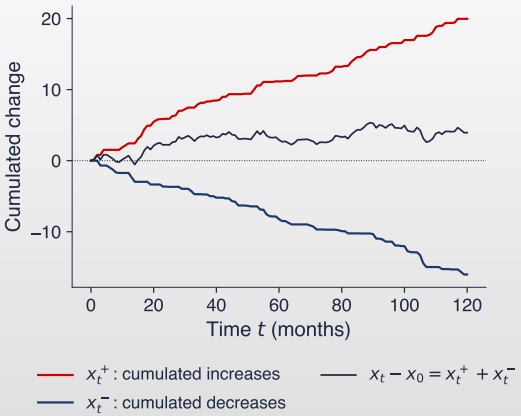
- Bounds test Pesaran, Shin and Smith (2001) (PSS: Pesaran, Shin and Smith): H_0: no level relationship, $\pi_{yy} = 0$ and $\pi_{yx} = 0$
  - F: the Wald statistic of the $1 + k$ restrictions divided by $1 + k$ (formula in the frame on Wald tests)
  - $t = \hat{\pi}_{yy}/\mathrm{se}(\hat{\pi}_{yy})$: the t-ratio of the adjustment coefficient alone
- Non-standard null distributions with two bounds
  - lower bound: the critical value if all x are I(0); upper bound: if all x are I(1)
  - cases I–V as for Johansen; the bounds depend on the case, on k and, in small samples, on T Narayan (2005)
- Assumptions
  - x_t weakly exogenous; each x is I(0) or I(1), never I(2); u_t serially uncorrelated

Prerequisites for Today (19/26): reading the bounds



- Simulated null densities of F, case III, $k = 1$, $T = 500$
  - the 95% quantiles are the bounds: 4.91 (x I(0)) and 5.81 (x I(1))
- Decision
  - F below the lower bound: no level relation
  - between the bounds: inconclusive (the order of x matters)
  - above the upper bound: a level relation, if t also rejects
- F rejects but t does not: the degenerate case, $\pi_{yx} \neq 0$ alone

Prerequisites for Today (20/26): NARDL: asymmetric adjustment



- Partial sums Shin, Yu and Greenwood-Nimmo (2014)

$$x_t^{\pm} = \sum_{j=\mathbf{1}}^{t} \Delta x_j^{\pm}$$

  - $\Delta x_j^+ = \max(\Delta x_j, 0)$, $\Delta x_j^- = \min(\Delta x_j, 0)$
  - $x_t = x_{\mathbf{0}} + x_t^+ + x_t^-$ (chart)
- NARDL (nonlinear ARDL): $x_{t-\mathbf{1}}$ replaced by $x_{t-\mathbf{1}}^+$, $x_{t-\mathbf{1}}^-$
  - coefficients π^+, π^-; long run $\theta^{\pm} = -\pi^{\pm}/\pi_{yy}$
  - impact coefficients $\omega_{\mathbf{0}}^+$, $\omega_{\mathbf{0}}^-$ of Δx_t^+, Δx_t^-
- Two symmetry hypotheses (Wald tests)
  - long run: $\theta^+ = \theta^-$; impact: $\omega_{\mathbf{0}}^+ = \omega_{\mathbf{0}}^-$

Prerequisites for Today (21/26): Wald tests

- Linear hypotheses H_0: $R\gamma = r_0$ on a coefficient vector γ

$$W = (R\hat{\gamma} - r_0)' \left(R\hat{V}R'\right)^{-1} (R\hat{\gamma} - r_0) \xrightarrow{d} \chi^2(q)$$

- Notation
  - $\hat{\gamma}$: the estimate; $\hat{V}$: its estimated covariance matrix; R ($q \times \dim\gamma$), r_0: known
  - q: the number of restrictions; $F = W/q$ is the F form used by the bounds test
- Reading
  - W measures the distance of $R\hat{\gamma}$ from r_0 in units of its own uncertainty
  - reject at 5% when W exceeds the 95% quantile of $\chi^2(q)$ (3.84 for $q = 1$)
  - NARDL long-run symmetry: $\theta^+ - \theta^-$ is nonlinear in γ, so R is replaced by its gradient (delta method)

Prerequisites for Today (22/26): cross-section dependence: the CD test

- Panel of N units over T periods; CD test Pesaran (2021)

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}$$

- Notation
  - $\hat{\rho}_{ij} = \sum_t e_{it} e_{jt} / \sqrt{\sum_t e_{it}^2 \sum_t e_{jt}^2}$: the correlation over time of the residuals (or demeaned series) e_{it} and e_{jt}
  - $N(N-1)/2$: the number of pairs $i < j$
- Reference distribution and reading
  - $CD \to N(0,1)$ under weak cross-section dependence; $|CD| > 1.96$ rejects at 5%
  - positive and negative $\hat{\rho}_{ij}$ can cancel: report also the mean $|\hat{\rho}_{ij}|$

Prerequisites for Today (23/26): panel unit-root tests

- ADF regression of unit i

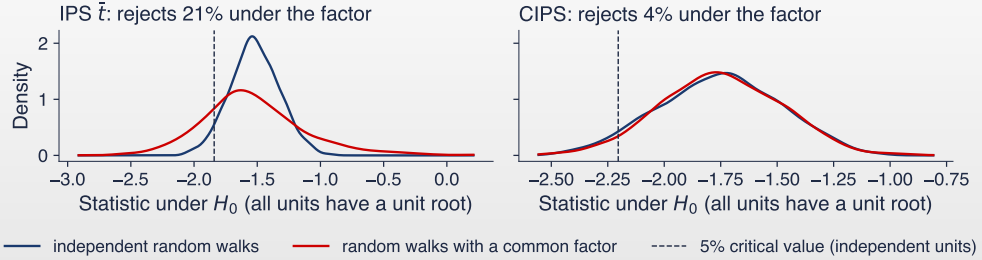
$$\Delta y_{it} = a_i + b_i y_{i,t-1} + \sum_{j=1}^{p_i} c_{ij} \Delta y_{i,t-j} + e_{it}$$

  - H_0: $b_i = 0$ for all i (all units have a unit root); t_i: the t-ratio of $\hat{b}_i$
- **LLC** Levin, Lin and Chu (2002): one common b; **IPS** Im, Pesaran and Shin (2003): heterogeneous b_i, $\bar{t} = N^{-1} \sum_i t_i$
  - IPS alternative: a non-zero fraction of the units is stationary, not all of them
  - both assume independent units: they over-reject under common shocks
- **CIPS** Pesaran (2007): cross-sectionally augmented ADF (CADF)

$$\Delta y_{it} = a_i + b_i y_{i,t-1} + c_i \bar{y}_{t-1} + d_i \Delta \bar{y}_t + e_{it}, \qquad CIPS = \frac{1}{N} \sum_{i=1}^{N} t_i^{CADF}$$

  - $\bar{y}_t = N^{-1} \sum_i y_{it}$: the cross-section average, a proxy for one common factor
  - critical values of IPS and CIPS: simulate independent random walks with the given N and T; reject for very negative values

Prerequisites for Today (24/26): IPS and CIPS under a common factor



- Simulated null: $N = 20$ random walks, $T = 30$, shocks $e_{it} + \lambda_i f_t$ with a common factor f_t and loadings $\lambda_i \in [0.5, 1.5]$
- With the 5% critical value of independent units
  - IPS rejects the true null in 21% of the panels: the common factor makes the t_i move together
  - CIPS keeps its size (4%): the averages $\bar{y}_{t-1}$, $\Delta\bar{y}_t$ absorb the factor

Prerequisites for Today (25/26): heterogeneous panel estimators

- Panel ARDL in error-correction form, unit i

$$\Delta y_{it} = \varphi_i (y_{i,t-1} - \theta_i' x_{i,t-1}) + \text{short-run terms}_i + e_{it}$$

  - $\varphi_i < 0$: the speed of adjustment; θ_i: the k long-run coefficients of unit i
- **MG** (mean group) Pesaran and Smith (1995): estimate each unit, then average

$$\hat{\theta}_{MG} = \frac{1}{N}\sum_{i=1}^{N} \hat{\theta}_i, \qquad \mathrm{SE}(\hat{\theta}_{MG}) = \frac{s}{\sqrt{N}}, \quad s^2 = \frac{1}{N-1}\sum_{i=1}^{N} (\hat{\theta}_i - \hat{\theta}_{MG})^2$$

  - consistent whether or not the long run is homogeneous, but imprecise when N is small
- **PMG** (pooled mean group) Pesaran, Shin and Smith (1999): $\theta_i = \theta$ for all i; φ_i and the short run free
  - efficient if the long run is homogeneous, inconsistent if it is not

Prerequisites for Today (26/26): the Hausman test

- Compare an estimator consistent under both hypotheses (MG) with one efficient under H_0 (PMG)

$$H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})'(V_{MG} - V_{PMG})^{-1}(\hat{\theta}_{MG} - \hat{\theta}_{PMG}) \xrightarrow{d} \chi^2(k)$$

- Notation
  - V_{MG}, V_{PMG}: the estimated covariance matrices; k: the number of long-run coefficients
  - $k = 1$: $H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})^2/(\mathrm{SE}^2_{MG} - \mathrm{SE}^2_{PMG})$
- Reading
  - H_0: long-run homogeneity; a large H rejects the common long run, so prefer MG
  - a small H: PMG is preferred, being more precise
  - under H_0, $V_{MG} - V_{PMG}$ is positive semi-definite, because PMG is efficient

A1: the trace test from eigenvalues [Solved]

Task

- $n = 3$, VAR(2) in levels, case 3, effective $T = 100$; eigenvalues $\hat{\lambda} = (0.30; 0.12; 0.03)$; 95% quantiles of the trace statistic for $n - r = 3, 2, 1$: 29.78, 15.48, 3.89.
- 1. Compute the trace statistics for $r = 0, 1, 2$ and the max-eigenvalue statistics.
- 2. Determine the rank by the sequential procedure.
- 3. Apply the Reinsel–Ahn factor and say whether the rank changes.
- Report: six statistics, one rank, one sentence.

Solution

- 1. $-100\ln(0.70) = 35.67$, $-100\ln(0.88) = 12.78$, $-100\ln(0.97) = 3.05$; trace: 51.50, 15.83, 3.05; max-eigenvalue: 35.67, 12.78, 3.05
- 2. $51.50 > 29.78$: reject $r = 0$; $15.83 > 15.48$: reject $r \leq 1$ (narrowly); $3.05 < 3.89$: $\hat{r} = 2$
- 3. Factor $(100 - 6)/100 = 0.94$: 48.41, 14.88, 2.86; now $14.88 < 15.48$, so $\hat{r} = 1$: the second relation does not survive the correction

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A2: degrees of freedom of LR tests [Proposed]

Solution discussed in class

Task

- $n = 4$ variables, case 3 ($n_1 = 4$), rank $r = 2$. Three hypotheses with their LR statistics. Model: A1.
- 1. Variable 4 is excluded from both relations, $\beta = H\varphi$ with H of dimension 4×3: give the degrees of freedom and decide at 5% for $LR = 3.1$.
- 2. Variable 1 is weakly exogenous: give the degrees of freedom and decide for $LR = 7.4$.
- 3. Both vectors are fully known from theory: give the degrees of freedom and decide for $LR = 9.0$.
- 4. How many restrictions just-identify β, and why do they not change the likelihood?
- Report: three degrees of freedom with decisions, one number, one sentence.

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A3: the Granger representation by hand [Solved]

Task

- Bivariate VECM without lags: $\Delta y_t = \alpha\beta' y_{t-1} + \varepsilon_t$, $\alpha = (-0.2, 0.1)'$, $\beta = (1, -1)'$, so $\Gamma = I$.
- 1. Find $\alpha_\perp$ and $\beta_\perp$ (first element normalised to 1).
- 2. Compute $C = \beta_\perp(\alpha_\perp'\beta_\perp)^{-1}\alpha_\perp'$ and check $\beta' C = 0$ and $C\alpha = 0$.
- 3. Write the common trend and explain which variable carries more of it.
- Report: two vectors, one matrix, two checks, one sentence.

Solution

- 1. $\alpha'\alpha_\perp = 0$: $-0.2a + 0.1b = 0$, $\alpha_\perp = (1, 2)'$; $\beta_\perp = (1, 1)'$
- 2. $\alpha_\perp'\beta_\perp = 3$, $C = \frac{1}{3}\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$; $\beta' C = (0, 0)$, $C\alpha = \frac{1}{3}(-0.2 + 0.2)(1, 1)' = 0$
- 3. Trend $\frac{1}{3}\sum_s(\varepsilon_{1s} + 2\varepsilon_{2s})$, common to both: y_2 adjusts less ($|\alpha_2| < |\alpha_1|$), so its shocks weigh twice as much in the trend

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A4: the permanent shock of a common-trends model [Proposed]

Solution discussed in class

Task

- $n = 3$, $r = 2$, $\alpha = \begin{pmatrix} -0.1 & 0 \\ 0 & -0.2 \\ 0.05 & 0.1 \end{pmatrix}$, $\Omega = \begin{pmatrix} 1 & 0.3 & 0.5 \\ 0.3 & 4 & 1 \\ 0.5 & 1 & 1 \end{pmatrix}$. Model: A3.
- 1. Find $\alpha_\perp$ (first element 1).
- 2. Compute $\alpha_\perp' \Omega \alpha_\perp$ and the impact vector b^P of the permanent shock.
- 3. Compute the share of the permanent shock in the one-step forecast error variance of each variable, $(b_i^P)^2/\Omega_{ii}$, and interpret.
- Report: one vector, one number, one vector, three shares.

A5: an ARDL in error-correction form [Solved]

Task

- ARDL(1, 1): $y_t = 0.5 + 0.7y_{t-\mathbf{1}} + 0.2x_t + 0.1x_{t-\mathbf{1}} + u_t$.
- 1. Rewrite it as a conditional ECM; give π_{yy}, π_{yx} and the long-run coefficient θ.
- 2. Compute the half-life of a disequilibrium.
- 3. Compute the cumulative multipliers of a permanent unit rise of x after 0, 1, 2 and 3 periods.
- Report: one equation, three numbers, one number, four numbers.

Solution

- 1. $\Delta y_t = 0.5 - 0.3y_{t-\mathbf{1}} + 0.3x_{t-\mathbf{1}} + 0.2\Delta x_t + u_t = 0.5 - 0.3(y_{t-\mathbf{1}} - x_{t-\mathbf{1}}) + 0.2\Delta x_t + u_t$; $\theta = 0.3/0.3 = 1$, long-run intercept $0.5/0.3 = 1.667$
- 2. $\ln 0.5 / \ln 0.7 = 1.94$ periods
- 3. $m_{\mathbf{0}} = 0.2$; $m_h = 0.7m_{h-\mathbf{1}} + 0.3$: 0.200, 0.440, 0.608, 0.726 $\rightarrow \theta = 1$

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A6: a bounds-test decision [Proposed]

Solution discussed in class

Task

- Case III, $k = 1$, $T = 60$ annual observations. 5% bounds: asymptotic [4.97; 5.71], simulated for $T = 60$ [5.18; 5.96]; t-bounds for $T = 60$ [-2.87; -3.26]. Model: A5.
- 1. For three studies with $F = 5.4$, $F = 5.8$ and $F = 6.1$, give the decision with the asymptotic and with the small-sample bounds.
- 2. In the study with $F = 6.1$, $t = -2.9$. What do you conclude?
- 3. Which study changes its conclusion when the correct bounds are used?
- Report: six decisions, one sentence, one study.

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A7: the CD statistic by hand [Solved]

Task

- $N = 4$ countries, $T = 25$; pairwise residual correlations $\hat{\rho}_{12} = 0.6$, $\hat{\rho}_{13} = 0.5$, $\hat{\rho}_{14} = 0.4$, $\hat{\rho}_{23} = 0.7$, $\hat{\rho}_{24} = 0.3$, $\hat{\rho}_{34} = 0.5$.
- 1. Compute the CD statistic and its decision at 5%.
- 2. Explain what a CD statistic close to zero would and would not prove.
- 3. Say which panel unit-root test is appropriate after this result.
- Report: one statistic, two sentences, one test.

Solution

- 1. $\sum \hat{\rho}_{ij} = 3.0$; $CD = \sqrt{2 \cdot 25/(4 \cdot 3)} \cdot 3.0 = 6.12 > 1.96$: reject weak dependence
- 2. Near zero: no average correlation; it does not prove independence, because positive and negative $\hat{\rho}_{ij}$ can cancel (report the mean $|\hat{\rho}_{ij}|$ too)
- 3. CIPS (cross-sectionally augmented), not LLC or IPS

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A8: the Hausman test of PMG against MG [Proposed]

Solution discussed in class

Task

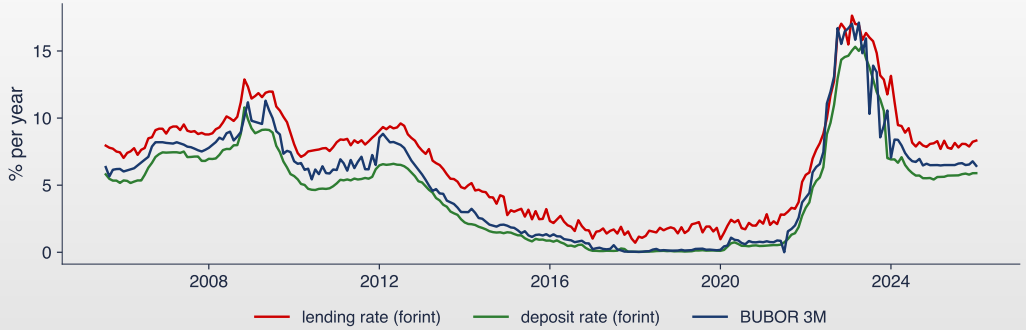
- Long-run income elasticity: MG 0.85 (SE 0.10), PMG 0.95 (SE 0.04). Four country estimates: 0.6, 0.9, 1.0, 0.9. Model: A7.
- 1. Compute the Hausman statistic and its p-value.
- 2. Compute the MG estimate and its standard error from the four country estimates.
- 3. Explain why $V_{MG} - V_{PMG}$ must be positive and what you would do if it were not.
- Report: two numbers, two numbers, two sentences.

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B1: the rank of the Hungarian pass-through system [Solved]

- **Question**: do Hungarian bank lending and deposit rates share long-run relations with the money-market rate?
- **Inputs**: lending and deposit rates in forint (IMF IFS), BUBOR 3M (Eurostat; isolated missing months interpolated), monthly, $T = 246$; case 2
- **Tasks**
  1. Select the lag order of the VAR in levels by AIC, BIC and HQ (up to 8).
  2. Compute the trace statistics with the BIC lag order and compare them with the simulated 95% quantiles of case 2.
  3. Apply the Reinsel–Ahn correction and compute wild-bootstrap p-values with 399 samples.
  4. Interpretation: is the rank robust to the small-sample corrections?
- **Report**: three lag orders, a table of three rows, one rank, one sentence
- Notebook: section B1

B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

$H(r)$	LR_{tr}	RA	cv 5%	wild p
$r = 0$	81.4	77.4	35.2	0.003
$r \leq 1$	31.2	29.6	20.5	0.005
$r \leq 2$	3.9	3.7	9.3	0.390

- Lags: AIC 8, BIC 4, HQ 4; $p = 4$; eigenvalues 0.188, 0.106, 0.016; all three methods give $\hat{r} = 2$
- Interpretation: the Hungarian rank survives both small-sample corrections
  - two cointegrating relations and one common trend
  - with a shorter sample or more lags the corrections can overturn the asymptotic rank: report all three

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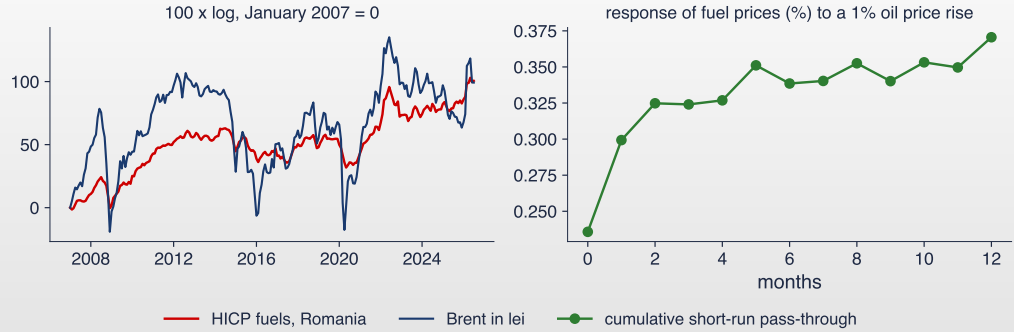
B2: tests on β and α for Hungary [Proposed]

- **Question**: is the long-run pass-through to Hungarian lending and deposit rates complete, and does BUBOR adjust to bank rates?
- **Inputs**: the data and the VECM of B1 with $r = 2$, case 2; model: B1
- **Tasks**
  1. Just-identify β: the lending relation excludes the deposit rate, the deposit relation excludes the lending rate; report both relations.
  2. Test complete pass-through to both rates (coefficient -1 on BUBOR in both vectors), then to the lending rate alone; give the degrees of freedom.
  3. Test weak exogeneity of BUBOR (a zero row of α) and report $\hat{\alpha}$.
  4. Interpretation: can a single-equation ARDL for the lending rate be used here?
- **Report**: two relations, three LR tests with degrees of freedom and p-values, one matrix, one sentence
- Notebook: section B2

B3: oil prices and Romanian fuel prices, bounds test [Solved]

- **Question**: is there a long-run relation between Romanian retail fuel prices and the oil price in lei?
- **Inputs**: y: $100 \ln$ HICP fuels and lubricants (CP0722, Eurostat); x: $100 \ln$ (Brent in USD, FRED $\times$ USD/RON, BNR monthly average); monthly, $T = 235$
- **Tasks**
  1. Select an ARDL by BIC (up to 6 lags each) in case III and estimate the conditional ECM.
  2. Compute the bounds F and t and compare them with asymptotic and small-sample 5% bounds.
  3. Repeat in case V (trend) and report the long-run elasticity in both cases.
  4. Interpretation: what does the absence of a level relation mean for fuel prices?
- **Report**: two models, two F and two t with bounds, two elasticities, one sentence
- Notebook: section B3

B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

- Case III, ARDL(1, 2): $F = 1.79$ against [4.97; 5.71] (asymptotic) and [5.06; 5.86] (T of the sample); $t = -1.27$ against $[-2.86; -3.22]$; $\hat{\theta} = 1.07$ (SE 0.54)
- Case V: $F = 4.05$ against [6.52; 7.41]; $t = -2.49$ against $[-3.41; -3.69]$; $\hat{\theta} = 0.51$ (SE 0.10)
- Short run (right panel): a 1% rise in the oil price raises fuel prices by 0.24% in the same month and by 0.37% after 12 months
- Interpretation: F is below the lower bound in both cases, so there is no stable level relation
  - excise duties, VAT changes and the 2022 price caps shift the retail price independently of oil
  - $\hat{\theta}$ is therefore not interpretable as a long-run elasticity

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B4: the KPSW common trend on longer samples [Proposed]

- **Question**: does the share of the permanent (balanced-growth) shock in US output fluctuations survive a longer sample and a trend in the great ratios?
- **Inputs**: 100 ln real per capita consumption of nondurables and services, fixed investment and GDP (FRED), quarterly, from 1949Q1; model: B1 and A4
- **Tasks**
  1. For samples ending in 1988Q4, 2019Q4 and 2025Q4, test the great ratios $\beta = ((1, 0, -1)', (0, 1, -1)')$ in case 3 and, with a restricted trend, in case 4.
  2. Impose the great ratios and compute the share of the permanent shock in the forecast error variance of output at 1, 8 and 24 quarters, in both cases.
  3. Interpretation: is the importance of trend shocks a property of the data or of the specification?
- **Report**: a table of six rows (sample and case) with an LR test and three shares, one sentence
- Notebook: section B4

Back Exercise list

B5: EU-27 inflation, IPS against CIPS [Proposed]

- **Question**: is annual inflation stationary in the EU-27 once the common shocks are taken into account?
- **Inputs**: annual HICP inflation, Eurostat prc_hicp_aind, $N = 27$, 1999–2025 (balanced); model: A7 and B1
- **Tasks**
  1. Compute the CD statistic of the demeaned series and the mean absolute correlation.
  2. Compute IPS $\bar{t}$ and CIPS with one lag and a constant; obtain 5% critical values and p-values from 1,000 simulations of independent random walks with the same N and T.
  3. Repeat from 2000 onward.
  4. Interpretation: does cross-section dependence change the conclusion here?
- **Report**: a table of two rows with CD, mean $|\rho|$, IPS and CIPS with p-values, one sentence
- Notebook: section B5

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C1: rockets and feathers in Romanian fuel prices [Proposed]

- **Question**: do Romanian fuel prices react more to oil-price increases than to decreases?
- **Inputs**: the data of B3; partial sums x_t^+, x_t^- of the changes of the oil price in lei; model: B3
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, variables, case, lag rule, the two asymmetry tests.
  2. Estimate the NARDL in case III with lags by BIC (up to 4); report the bounds F with $k = 2$ and the long-run coefficients θ^+, θ^-.
  3. Test long-run symmetry and impact symmetry by Wald tests.
  4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- **Report**: a five-line plan, one F, two coefficients, two Wald tests, a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise cointegration testing. The answer:
- (a) For two interest rates without drift, use the case-3 critical value 3.84 for H(1).
- (b) The eigenvalues of the Johansen problem are squared canonical correlations.
- (c) With n = 4 and r = 2, testing two fully known cointegrating vectors has 2 degrees of freedom.
- (d) ARDL bounds tests are valid whatever the integration order of the regressors, including I(2).
- (e) A weakly exogenous variable has a zero row in alpha, so it can be conditioned on.
- (f) If the IPS test rejects a panel unit root, every country is stationary.
- (g) The bootstrap rank test should resample from the unrestricted VAR in levels.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- The rank comes from eigenvalues, but its critical values depend on the deterministic case and on the sample size
- Count the degrees of freedom of every test on β and α by hand
- An ARDL needs weak exogeneity and correct bounds; the absence of a level relation is a result too
- In panels, measure the cross-section dependence before trusting a unit-root or cointegration test
- An AI answer is a draft: check the case, the degrees of freedom, the alternative hypothesis and the bootstrap design

After the seminar

- Lecture 4 develops each topic of today: the likelihood behind Johansen, the five cases, the bootstrap, identification, common trends, ARDL, panels
- C1 can grow into a team project: asymmetric price transmission with weekly data
- Reading: Johansen (1991); Pesaran, Shin and Smith (2001); Pesaran, Shin and Smith (1999); Pesaran (2007)
- **The seminar is for practice and is not graded**

References (1/2)

- Bacon, R. W. (1991). Rockets and feathers: The asymmetric speed of adjustment of UK retail gasoline prices to cost changes. *Energy Economics*, 13(3), 211–218. doi:10.1016/0140-9883(91)90022-R
- Cavaliere, G., Rahbek, A., & Taylor, A. M. R. (2012). Bootstrap determination of the co-integration rank in vector autoregressive models. *Econometrica*, 80(4), 1721–1740. doi:10.3982/ECTA9099
- Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276. doi:10.2307/1913236
- Im, K. S., Pesaran, M. H., & Shin, Y. (2003). Testing for unit roots in heterogeneous panels. *Journal of Econometrics*, 115(1), 53–74. doi:10.1016/S0304-4076(03)00092-7
- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580. doi:10.2307/2938278
- Johansen, S. (1995a). *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press. doi:10.1093/0198774508.001.0001
- King, R. G., Plosser, C. I., Stock, J. H., & Watson, M. W. (1991). Stochastic trends and economic fluctuations. *American Economic Review*, 81(4), 819–840. jstor.org
- Kripfganz, S., & Schneider, D. C. (2020). Response surface regressions for critical value bounds and approximate p-values in equilibrium correction models. *Oxford Bulletin of Economics and Statistics*, 82(6), 1456–1481. doi:10.1111/obes.12377
- Levin, A., Lin, C.-F., & Chu, C.-S. J. (2002). Unit root tests in panel data: Asymptotic and finite-sample properties. *Journal of Econometrics*, 108(1), 1–24. doi:10.1016/S0304-4076(01)00098-7

References (2/2)

- MacKinnon, J. G., Haug, A. A., & Michelis, L. (1999). Numerical distribution functions of likelihood ratio tests for cointegration. *Journal of Applied Econometrics*, 14(5), 563–577. doi:10.1002/(SICI)1099-1255(199909/10)14:5<563::AID-JAE530>3.0.CO;2-R
- Narayan, P. K. (2005). The saving and investment nexus for China: Evidence from cointegration tests. *Applied Economics*, 37(17), 1979–1990. doi:10.1080/00036840500278103
- Pesaran, M. H. (2007). A simple panel unit root test in the presence of cross-section dependence. *Journal of Applied Econometrics*, 22(2), 265–312. doi:10.1002/jae.951
- Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*, 60(1), 13–50. doi:10.1007/s00181-020-01875-7
- Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of Applied Econometrics*, 16(3), 289–326. doi:10.1002/jae.616
- Pesaran, M. H., Shin, Y., & Smith, R. P. (1999). Pooled mean group estimation of dynamic heterogeneous panels. *Journal of the American Statistical Association*, 94(446), 621–634. doi:10.1080/01621459.1999.10474156
- Pesaran, M. H., & Smith, R. (1995). Estimating long-run relationships from dynamic heterogeneous panels. *Journal of Econometrics*, 68(1), 79–113. doi:10.1016/0304-4076(94)01644-F
- Reinsel, G. C., & Ahn, S. K. (1992). Vector autoregressive models with unit roots and reduced rank structure: Estimation, likelihood ratio test, and forecasting. *Journal of Time Series Analysis*, 13(4), 353–375. doi:10.1111/j.1467-9892.1992.tb00113.x
- Shin, Y., Yu, B., & Greenwood-Nimmo, M. (2014). Modelling asymmetric cointegration and dynamic multipliers in a nonlinear ARDL framework. In R. C. Sickles & W. C. Horrace (Eds.), *Festschrift in Honor of Peter Schmidt* (pp. 281–314). Springer. doi:10.1007/978-1-4899-8008-3_9